

Design and Manufacture of an RC Aircraft

Overview:

This project comprised the complete conceptual design, aerodynamic analysis and physical manufacture of a fixed-wing RC aircraft, tasked with carrying a payload while remaining stable and controllable using only electric propulsion. The aircraft had to be modular and fit into a cardboard transportation box, sized at 1200x660x433 mm.

The aircraft disassembled in the cardboard box is shown below:



Figure 11: Unfinished RC Aircraft disassembled in transportation box

An aircraft technical log was maintained, with the wing, tailplane, stability and performance parameters calculated. The aerodynamics were verified in ANSYS Fluent and the geometry was developed in SOLIDWORKS CAD before being manufactured using hot-wire-cut foam and plywood.

An initial cargo-style airframe was chosen, as the strategy was to prioritise lifting capacity. However, this later had to be scrapped due to the limitations of the specified electric propulsion system and material yield strength. The design focus shifted towards a smaller aircraft prioritising aerodynamic efficiency over payload volume.

The airfoil selected was a standard unmodified Clark Y. This airfoil was selected for its favourable lift-to-drag ratio and simple manufacturability. The aircraft was driven by a 12x6 inch propeller using an E-Flite Power 10 1100 Kv motor and a single three-cell (2200 mAh) lithium-polymer battery. MTOW was 1.6 kg with 350 g allocated for payload.



Figure 12: Finished RC Aircraft undergoing bench testing

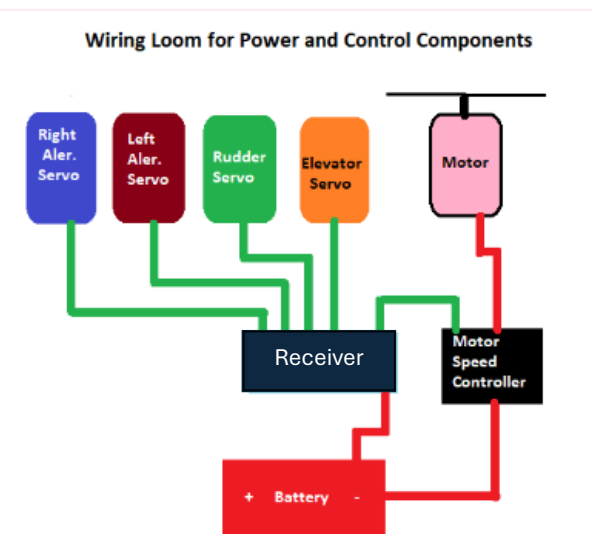


Figure 13: Wiring Loom for Power and Control Components

Longitudinal stability was addressed through the sizing of the components and the placement of the centre of gravity. The neutral point was located at 45.3% of the MAC aft of the leading edge, with the loaded centre of gravity at ~25% of the chord. This resulted in a static margin of ~0.2. A tailplane setting angle of 2.99° relative to the horizontal fuselage datum was chosen to trim the aircraft in cruise, accounting for a downwash gradient of 0.53.

The centre of gravity was subsequently verified experimentally using a three-point weighing strategy, with the measured centre of gravity aligning with the predicted centre of gravity derived from the SOLIDWORKS model.

The wing was sized to a gross area of 0.275m² at an aspect ratio of 4.4, resulting in a MAC of 0.25m. The lift-curve slope for the 3D Clark Y was calculated to be 4.19 per radian, and the maximum lift coefficient was 1.39. The Reynolds number was 1.95x10⁵ and Oswald efficiency factor was obtained as 0.921.

Table 2: Velocity Performance Parameters of Aircraft

Performance Parameter	Value (m/s)
Stall Speed	8.56
Take-Off Speed	10.27
Cruise Speed	11.12
Maximum Speed	13.82

Two-dimensional CFD analysis was performed in ANSYS Fluent to determine the aerodynamic characteristics. The determined C_L and C_D (Coefficient of Lift and Drag respectively) aligned with the values in the technical log, verifying the turbulence model and meshing strategy.

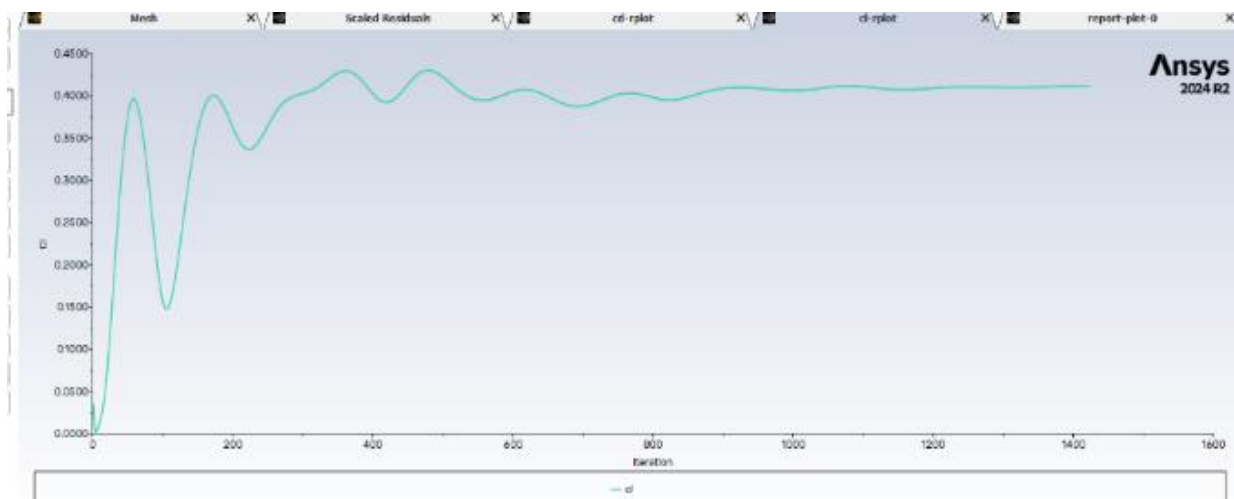


Figure 14: CL of Airfoil

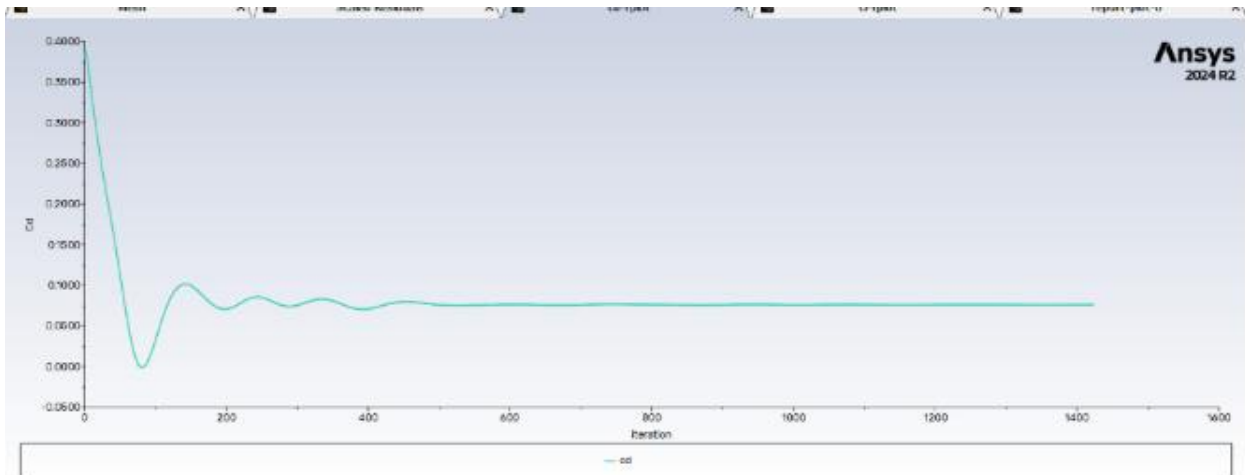


Figure 15: CD of Airfoil

The aircraft was manufactured principally from 5mm foamboard, with the wing and control surfaces hot-wire-cut from foam and plywood supports laser cut from DXF files. The fuselage assembly was joined using riveted joints, which improved structural robustness when compared to joining individual panels.

A payload compartment of four interlocking foamboard slots was designed to retain the cargo (tennis balls) and restrict payload movement. The nose-leg mounts were 3D-printed to suit the airframe, while a proprietary main undercarriage was adopted. The foam airfoil was reinforced with a plywood spar to maximise structural integrity while minimising added weight. The control system was wired with the aileron servo leading to a channel cut into the underside of the wing to preserve a clean aerodynamic surface, which can be seen in **Figure 11**. Servo function and motor response were verified with a bench test, ensuring the aircraft was ready for test flight. The aircraft underwent a successful test flight with the payload despite considerable turbulence arising from weather conditions.

Technical drawings were prepared outlining the BOM and the manufacturing techniques required to assemble the aircraft.

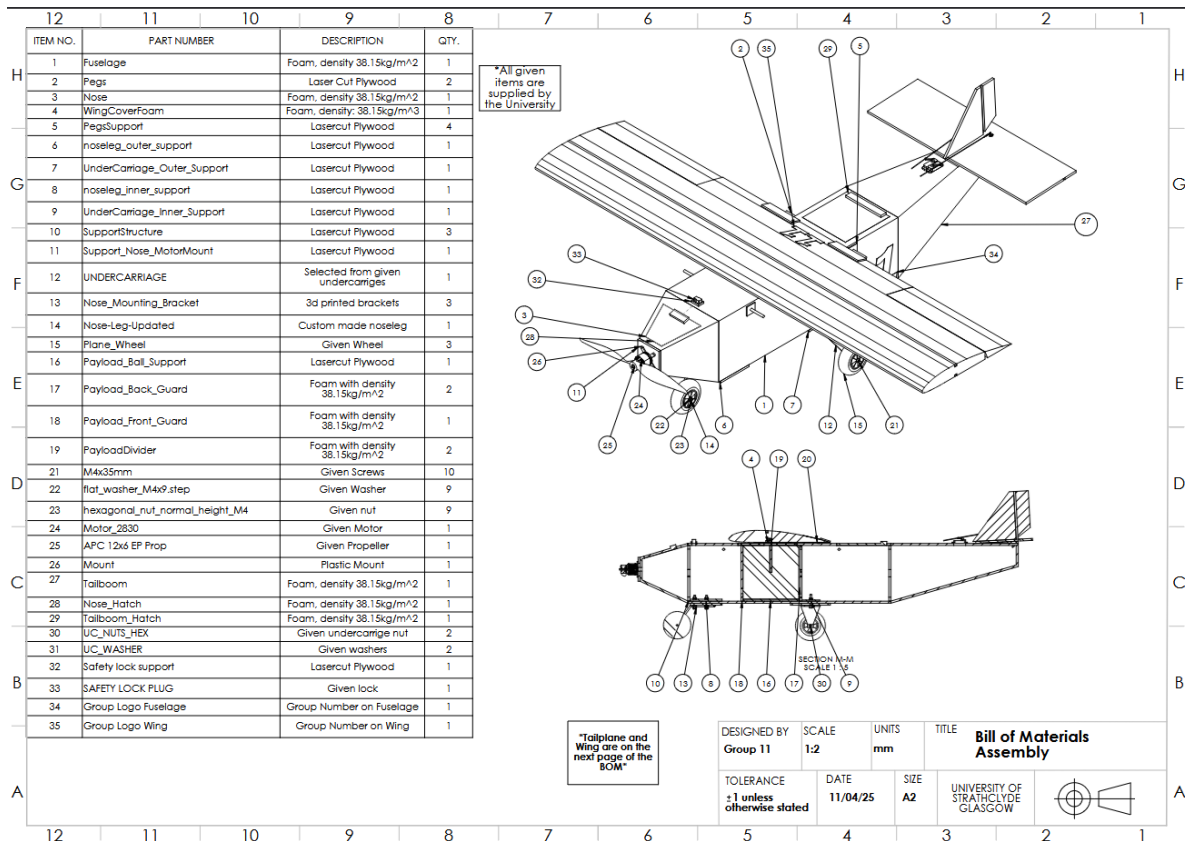


Figure 16: BOM of Aircraft

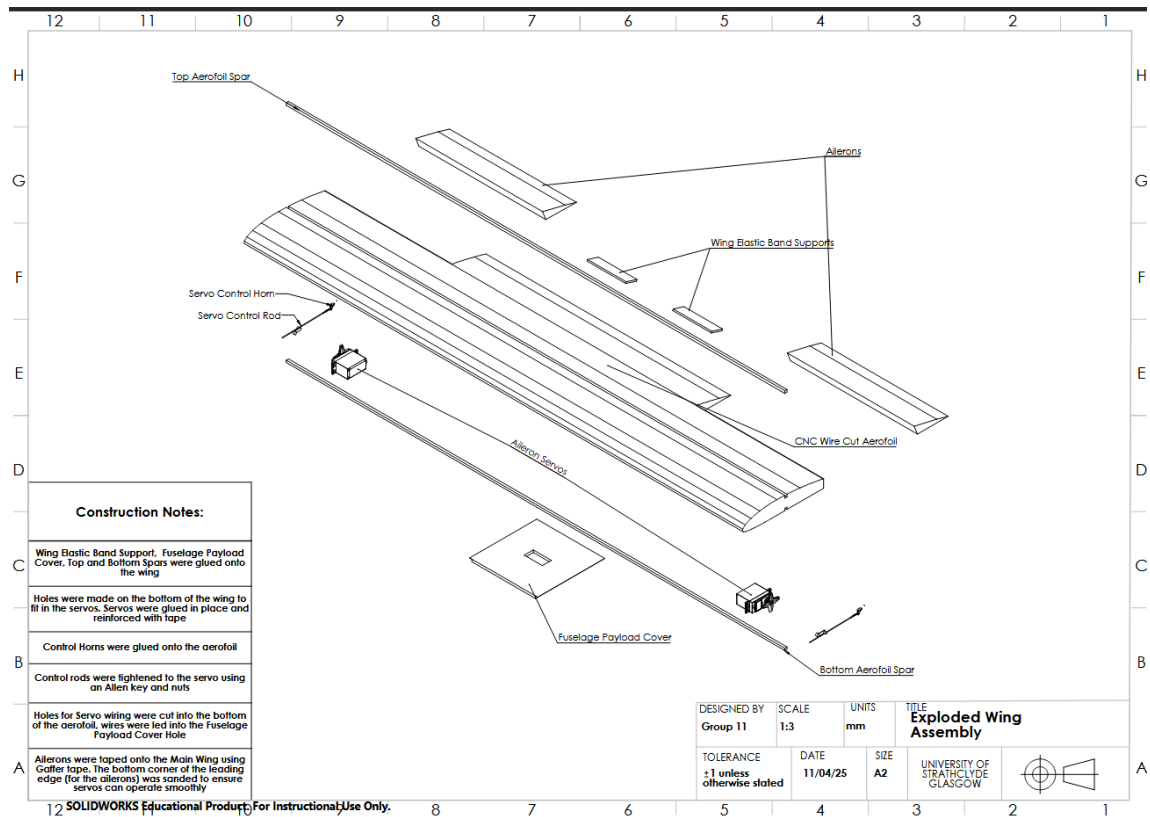


Figure 17: Wing Assembly Drawing